

Device Names

Device names are connected to the operating system through Kernel modules, and have the following general parameters, as seen below.

`/dev/xxyn`, where:

- **xx** is the type of device (**sd** = scsi, **hd** = ide, **vd** = virtual disk, etc.)
- **y** is a letter indicating the discovery order (**sda** = first, **sdb** = second, etc.)
- **n** is the partition number ('missing' = whole disk, **1** = first partition, **2** = second partition, etc.)

You can see some examples of filesystem device identifiers in the table below.

Table 5.2: List of Device Names and Their Device Types

DEVICE NAME	DEVICE TYPE
<code>/dev/fd0 ...</code> <code>/dev/fd1</code>	Floppy disk, first and second
<code>/dev/sda ...</code> <code>/dev/sdb</code>	Device using the SCSI driver (e.g., USB), first and second
<code>/dev/nvme0n1p1</code>	NVMe SSD (Non-Volatile Memory extended Solid State Disk). For more details, check out the official NVMe Express website
<code>/dev/vda ...</code> <code>/dev/vdb</code>	Virtual disk
<code>/dev/xvda</code>	Xen virtual disk

Most devices are described in the [*Linux Kernel User's and Administrator's Guide*](#).

As noted, the disk name is assigned at initialization time. The system usually assigns the same name to the same device because the device is discovered in the same sequence each boot. However, if a change occurs with the number of devices, the name may be different.

Filesystem labels are used to easily identify and assign filesystems. They can be up to 16 characters long. One disadvantage with using filesystem labels is that there is no "duplicate" checking, which can lead to confusion. There is another option to positively identify a filesystem, using the UUID (Universally Unique Identifier), a 128-bit number with a high probability of uniqueness in time and space.

The screenshot below shows multiple disks with partitions, labels, UUID's and [mount points](#).

```

root@grumpy:~
File Edit View Search Terminal Help
[root@grumpy ~]# lsblk -f
NAME        FSTYPE     LABEL      UUID                                 MOUNTPOINT
sda
├─sda1      vfat       SYSTEM     E0E3-1B3E
├─sda2
├─sda3      ntfs       Windows    44A0E845A0E83F54
├─sda4      ntfs       WinRE_DRV  56D8E94CD8E92B47
├─sda5      ext4       fc30-boot  7d528719-b5d8-4897-a4d6-7a274ba4567f
└─sda6      ext4       data_2     fddc4e8b-4727-41dd-9665-b31a3f49cdd7
sdb
├─sdb1      ext4       LF-VMs     835895f4-1117-4f91-8579-3191cf5ecc17  /home/LF-VMs
├─sdb2      ext4       lee-home   02fbb434-695f-43fe-95bd-81f5aa962172  /home/lee
└─sdb3      ext4       LF-Images  503d5cdb-02da-491d-a56e-e08494b10dec   /var/lib/libvirt/images
nvme0n1
├─nvme0n1p1 vfat       B0BD-513C  /boot/efi
├─nvme0n1p2 ext4       140711e1-522c-4216-8be0-b4915fc18eed  /boot
└─nvme0n1p3 LVM2_member
   ├─cl-root xfs       47bd3769-b004-4c01-9026-9b42fb880579  /
   ├─cl-swap swap      7c3957ae-3cb6-4723-995d-0aea0efa9722  [SWAP]
   └─cl-home xfs       0e495137-c739-4c5f-afaa-87e66d976650  /home
[root@grumpy ~]#

```

Sample Information about Filesystem